KAIZEN MATHS

Lesson starter

Further Vectors: Lines, Planes and Products

Scalar Product

Name _____

S1

Use the scalar product to find each angle.

Find the angle between a = $\begin{pmatrix}1\\2\\2\end{pmatrix}$ and b = $\begin{pmatrix}4\\1\\0\end{pmatrix}$.

S2

Use the scalar product to find each parameter.

a = $\begin{pmatrix}3\\-1\\3\end{pmatrix}$ and b = $\begin{pmatrix}1\\3\\-3\end{pmatrix}$ + t $\begin{pmatrix}0\\1\\0\end{pmatrix}$. Find t if a is perpendicular to b.

S3

Find the vector projection requested.

Find the vector projection of a = $\begin{pmatrix}2\\3\\6\end{pmatrix}$ onto b = $\begin{pmatrix}1\\2\\2\end{pmatrix}$.

S4

Use the scalar product to find each angle.

Find the angle between a = $\begin{pmatrix}2\\3\\6\end{pmatrix}$ and b = $\begin{pmatrix}2\\2\\1\end{pmatrix}$.

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S4

Use the scalar product to find each angle.

Find the angle between a = $\begin{pmatrix}2\\3\\6\end{pmatrix}$ and b = $\begin{pmatrix}2\\2\\1\end{pmatrix}$.

ALG-ADB4-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Scalar Product

Name _____

EA1

Use the scalar product to find each parameter.

a = $\begin{pmatrix}1\\1\\3\end{pmatrix}$ and b = $\begin{pmatrix}3\\-1\\3\end{pmatrix}$ + t $\begin{pmatrix}0\\1\\0\end{pmatrix}$. Find t if a is perpendicular to b.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Scalar Product

Name _____

EB1

Find the vector projection requested.

Find the vector projection of a = $\begin{pmatrix}3\\4\\0\end{pmatrix}$ onto b = $\begin{pmatrix}1\\2\\2\end{pmatrix}$.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Scalar Product

Name _____

EC1

Use the scalar product to find each angle.

Find the angle between a = $\begin{pmatrix}2\\1\\2\end{pmatrix}$ and b = $\begin{pmatrix}3\\2\\1\end{pmatrix}$.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Scalar Product

Name _____

EA1

Use the scalar product to find each parameter.

a = $\begin{pmatrix}1\\1\\3\end{pmatrix}$ and b = $\begin{pmatrix}3\\-1\\3\end{pmatrix}$ + t $\begin{pmatrix}0\\1\\0\end{pmatrix}$. Find t if a is perpendicular to b.

Show your working

Teacher answers & working

Further Vectors: Lines, Planes and Products · Scalar Product

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S1

Use the scalar product to find each angle.

Find the angle between a = $\begin{pmatrix}1\\2\\2\end{pmatrix}$ and b = $\begin{pmatrix}4\\1\\0\end{pmatrix}$.

Answer:

$\theta \approx 61.0°$

Working:

Use $a \cdot b = |a||b| \cos\theta$.

$a \cdot b = (1)(4) + (2)(1) + (2)(0) = 6$

$|a| = 3$ and $|b| = \sqrt{17}$

$\cos\theta = \frac{6}{3\times\sqrt{17}}$

$\theta \approx 61.0°$

S2

Use the scalar product to find each parameter.

a = $\begin{pmatrix}3\\-1\\3\end{pmatrix}$ and b = $\begin{pmatrix}1\\3\\-3\end{pmatrix}$ + t $\begin{pmatrix}0\\1\\0\end{pmatrix}$. Find t if a is perpendicular to b.

Answer:

t = -9

Working:

If two vectors are perpendicular, their scalar product is 0.

b = $\begin{pmatrix}1\\3+t\\-3\end{pmatrix}$

$a \cdot b = 3(1) + -1(3 + t) + 3(-3)$

$-9 - t = 0$

$-t = 9$

$t = -9$

Teacher answers & working

Further Vectors: Lines, Planes and Products · Scalar Product

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S3

Find the vector projection requested.

Find the vector projection of $a = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ onto $b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

Answer:

$\text{proj}_b a = \frac{20}{9}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$

Working:

Use $\text{proj}_b a = \frac{a \cdot b}{b \cdot b} b$.

$a \cdot b = 20$

$b \cdot b = 9$

$\text{proj}_b a = \frac{20}{9}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$

$\text{proj}_b a = \frac{20}{9}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$

S4

Use the scalar product to find each angle.

Find the angle between $a = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ and $b = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$.

Answer:

$\theta \approx 40.4°$

Working:

Use $a \cdot b = |a||b| \cos \theta$.

$a \cdot b = (2)(2) + (3)(2) + (6)(1) = 16$

$|a| = 7$ and $|b| = 3$

$\cos \theta = \frac{16}{7 \times 3}$

$\theta \approx 40.4°$

Teacher answers & working

Further Vectors: Lines, Planes and Products · Scalar Product

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EA1

Use the scalar product to find each parameter.

a = $\begin{pmatrix}1\\1\\3\end{pmatrix}$ and b = $\begin{pmatrix}3\\-1\\3\end{pmatrix}$ + t $\begin{pmatrix}0\\1\\0\end{pmatrix}$. Find t if a is perpendicular to b.

Answer:

t = -11

Working:

If two vectors are perpendicular, their scalar product is 0.

b = $\begin{pmatrix}3\\-1+t\\3\end{pmatrix}$

a · b = 1(3) + 1(-1 + t) + 3(3)

11 + t = 0

t = -11

t = -11

EB1

Find the vector projection requested.

Find the vector projection of a = $\begin{pmatrix}3\\4\\0\end{pmatrix}$ onto b = $\begin{pmatrix}1\\2\\2\end{pmatrix}$.

Answer:

$\text{proj}_b a = \frac{11}{9}\begin{pmatrix}1\\2\\2\end{pmatrix}$

Working:

Use $\text{proj}_b a = \frac{a \cdot b}{b \cdot b} b$.

a · b = 11

b · b = 9

$\text{proj}_b a = \frac{11}{9}\begin{pmatrix}1\\2\\2\end{pmatrix}$

$\text{proj}_b a = \frac{11}{9}\begin{pmatrix}1\\2\\2\end{pmatrix}$

Teacher answers & working

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EC1

Use the scalar product to find each angle.

Find the angle between $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$.

Answer:

$\theta \approx 27.0°$

Working:

Use $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$.

$\mathbf{a} \cdot \mathbf{b} = (2)(3) + (1)(2) + (2)(1) = 10$

$|\mathbf{a}| = 3$ and $|\mathbf{b}| = \sqrt{14}$

$\cos\theta = \frac{10}{3\times\sqrt{14}}$

$\theta \approx 27.0°$